

Experimental demonstration of long-distance continuous-variable quantum key distribution

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Distributing secret keys with information-theoretic security is arguably one of the most important achievements of the field of quantum information processing and communications¹. The rapid progress in this field has enabled quantum key distribution in real-world conditions^{2,3} and commercial devices are now readily available. Quantum key distribution systems based on continuous variables⁴ provide the major advantage that they only require standard telecommunication technology. However, to date, these systems have been considered unsuitable for long-distance communication^{5–7}. Here, we overcome all previous limitations and demonstrate for the first time continuous-variable quantum key distribution over 80 km of optical fibre. All aspects of a practical scenario are considered, including the use of finite-size data blocks for secret information computation and key distillation. Our results correspond to an implementation guaranteeing the strongest level of security for quantum key distribution reported so far for such long distances and pave the way to practical applications of secure quantum communications.

Long-distance experiments in quantum information science, and in particular featuring quantum key distribution (QKD), are of the utmost importance for future technological applications. Such experiments will allow the integration of quantum devices in current secure infrastructures and in future networks based on quantum repeaters⁸. The quest for long-distance QKD in recent years has led to several successful demonstrations^{9–12}; however, improving security guarantees and performance in practical conditions in these implementations remains an issue. These experiments use discrete-variable or distributed-phase-reference protocols¹, where key information is encoded on the properties of single photons. Alternatively, in continuous-variable (CV) QKD protocols, light carries continuous information such as the value of a quadrature component of a coherent state. Such protocols have been implemented in various situations^{5–7,13–17}. Their key feature is that dedicated photon-counting technology can be replaced by the homodyne detection techniques that are widely used in classical optical communications. However, CVQKD schemes require complex post-processing procedures, mostly related to error correction. These have limited their operation to less than 25 km of optical fibre^{5–7}, a communication span that may be insufficient for network cryptographic applications. Furthermore, implementations so far have been based on security proofs valid in the limit of infinitely large data blocks, but the finite size of real data must be taken into account according to state-of-the-art security analysis^{18,19}.

Here, we demonstrate the distribution of secret keys over a distance of 80 km, using continuous variables and security proofs

compatible with the use of finite-size data blocks for the generation of the secret key. This remarkable range improvement was made possible by system operation in a regime of signal-to-noise ratios (SNRs) between one and two orders of magnitude lower than in earlier implementations. It was previously overlooked that techniques developed for error correction of Gaussian signals can perform close to the optimal bounds for low SNR. Implementing such error-correction codes at high speed and in an optical environment featuring excellent stability, which enables the acquisition of large data blocks, allowed us to access previously inaccessible parameter regions. This was a key element for the present experiments.

In the experimental set-up, shown in Fig. 1, we implement the standard GG02 coherent-state CVQKD protocol²⁰. The sender, Alice, prepares coherent states with a Gaussian modulation and sends them to the receiver, Bob, who measures one of the quadratures with a homodyne detection system. A reverse reconciliation scheme is used in which Alice and Bob use Bob's data to establish the secret key¹³. In practice, coherent light pulses are generated by a 1,550 nm laser diode at a repetition rate of 1 MHz, and split into a weak signal and a strong local oscillator (LO), which provides the required phase reference. The signal pulses are then randomly modulated in both quadratures and transmitted together with the LO pulses in a single fibre using time- and polarization-multiplexing techniques. At the receiver side, the pulses interfere on a homodyne detector, the output of which is proportional to the signal quadrature selected by a phase modulator placed at Bob's LO path. Owing to the improved design with respect to previous implementations, the set-up presents excellent stability, ensured by several feedback controls (see Methods for details).

The security of the implemented protocol is well established against collective attacks, both in the asymptotic^{21,22} and finite-size regimes¹⁹. Moreover, collective attacks have been shown to be asymptotically optimal^{23,24}. Here, we consider security proofs pertaining to such attacks, also taking into account finite-size effects. The Gaussian modulation used in the implemented protocol maximizes the mutual information between Alice and Bob, thus offering an optimal theoretical key rate. However, it is hard to reconcile correlated Gaussian variables, especially at low SNRs, which are inherent in long-distance experiments. Indeed, the secure distance of previous demonstrations of fibre-based CVQKD^{5,7} was limited to 25 km because no efficient error-correction procedure was available at low SNRs. Here, we use a multidimensional reconciliation protocol²⁵, which transforms a channel with a Gaussian modulation into a virtual binary modulation channel, with a capacity loss that is very low at low SNR. This enables the use of error-correction codes designed for the binary input additive white Gaussian noise

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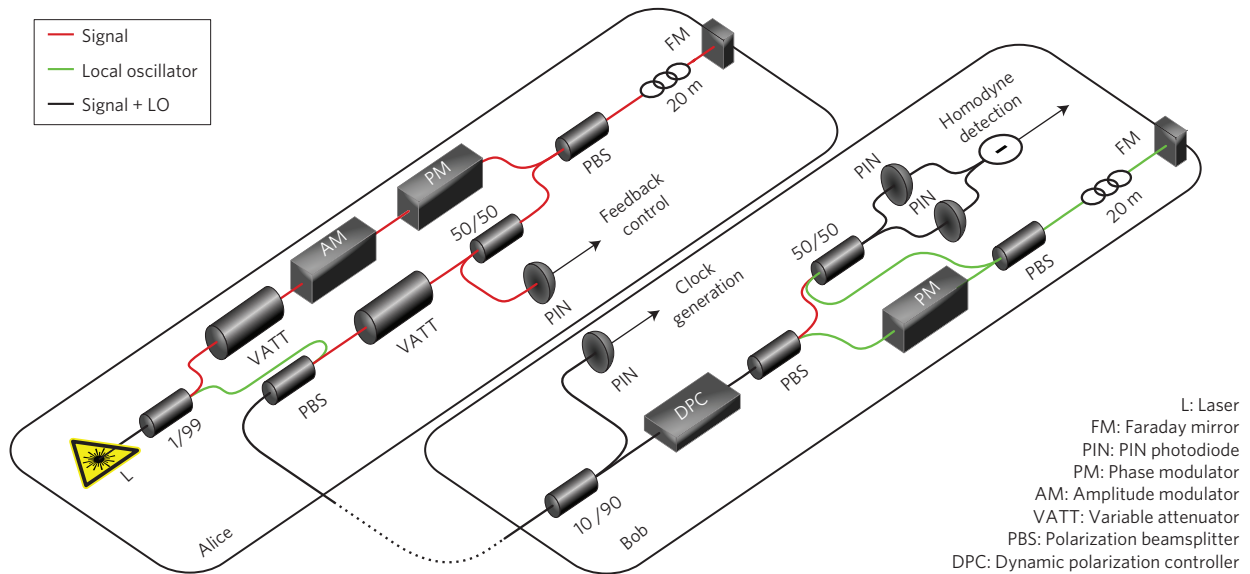


Figure 1 | Optical layout of the long-distance CVQKD prototype. Alice sends Bob 100 ns coherent light pulses generated by a 1,550 nm telecom laser diode pulsed with a frequency of 1 MHz. These pulses are split into a weak signal and a strong LO with an unbalanced coupler. The signal pulse is modulated with a centred Gaussian distribution using an amplitude and phase modulator. The variance is controlled using a coarse variable attenuator and the amplitude modulator. The signal pulse is 200 ns delayed with respect to the LO pulse using a 20 m delay line and a Faraday mirror. Both pulses are multiplexed with orthogonal polarization using a polarizing beamsplitter (PBS). The time- and polarization-multiplexed pulses are then sent through the channel. They are demultiplexed on Bob's side with another PBS combined with active polarization control. A second delay line on Bob's side allows for time superposition of the signal and LO pulses. After demultiplexing, the signal and LO interfere on a shot-noise-limited balanced pulsed homodyne detector. A phase modulator on the LO path allows for random choice of the measured signal quadrature. Alice and Bob are located in the same laboratory and separated by fibre spools. The maximum operating distance of the experiment is 80 km.

channel, which has typical efficiencies, for an arbitrarily low SNR, of 0.95 extracted bits per theoretically available bit²⁶. This leads to a significant extension of the secure distance.

Let us now look at the parameters that are relevant for the extraction of the secret key. As a result of the Gaussian optimality theorem^{21,22}, Alice and Bob's two-mode state at the output of the quantum channel is fully characterized by Alice's modulation variance V_A , channel transmission T and excess noise ξ , which is added by the channel. Both V_A and ξ are expressed in shot noise units. These parameters, together with the shot noise, are estimated in real time using a parameter estimation process, during which a fraction of the samples is randomly revealed. The other parameters used to compute an estimate of the secret information that can be extracted from the shared data, namely the electronic noise v_{el} and the efficiency of the homodyne detection η , are assumed not to be accessible to Eve and are measured during a secure calibration procedure that takes place before the deployment of the system. For simplicity, we make the standard assumption that Eve does not tamper with the local oscillator, but we emphasize that countermeasures against such tampering have been proposed (see Supplementary section, 'Local oscillator manipulation', for details). The modulation variance is adjusted in real time so as to be at all times as close as possible to the SNR corresponding to the threshold of an available code.

Privacy amplification allows extracting the secret information from the identical strings shared by Alice and Bob after error correction. In addition to the amount of data revealed during error correction, we compute an upper bound on the eavesdropper's information on the corrected string for collective attacks in both the asymptotic regime, where all the experimental parameters are assumed to be known with infinite precision, and in the finite-size regime, where the parameters are estimated over large data pulse sets. The stability of our system allows us to obtain a positive secret key rate at long distances in both regimes.

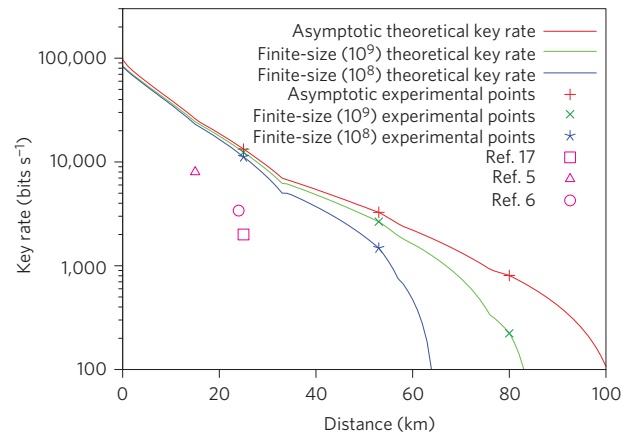


Figure 2 | Key rate produced by the system after error correction and privacy amplification over 24 h. There is a SNR of 1.1 on Bob's side at 25 km (5.0 dB losses), a SNR of 0.17 at 53 km (10.6 dB losses), and a SNR of 0.08 on Bob's side at 80.5 km (16.1 dB losses). In red, the rate is calculated assuming an eavesdropper is able to perform collective attacks in the asymptotic regime. This rate is also valid against arbitrary attacks. In green (blue), the rate is calculated assuming an eavesdropper is able to perform collective attacks taking into account finite-size effects with block size of 1×10^9 (1×10^8) and security parameter $\epsilon = 1 \times 10^{-10}$. The odd shape of the curves results from the use of a small set of error-correcting codes optimized to perform data reconciliation in specific ranges of SNR. The homodyne detection is characterized by an efficiency of $\eta = 0.552$, known with an uncertainty of $\Delta\eta = 0.025$, and an electronic noise variance $v_{el} = 0.015$, known with an uncertainty of $\Delta v_{el} = 0.002$. For comparison, previous state-of-the-art experimental results are shown^{5,6,17}; these are all restricted to distances below 25 km and do not take finite-size effects into account.

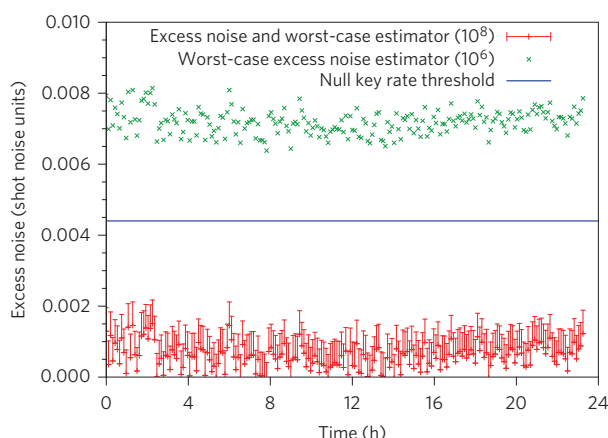


Figure 3 | Experimental excess noise measured over 24 h with a SNR of 0.17 on Bob's side. For this measurement we used 53 km of standard optical fibre corresponding to 10.6 dB losses. The red plus symbols correspond to measurements performed on blocks of size 1×10^8 , each corresponding to roughly 6 min of data acquisition, and indicate the experimentally measured excess noise (lower point) as well as the worst-case estimator for the excess noise (upper point) compatible with this data. The probability that the true value of the excess noise is underestimated by the estimator because of statistical fluctuations is less than 1×10^{-10} , assuming that the noise is Gaussian. This worst-case estimator is the value used to compute the secret key rate when finite-size effects are taken into account. For comparison, the green crosses correspond to the worst-case estimator if the estimation were performed on blocks of size 1×10^6 . The blue line indicates the maximal value of excess noise that allows for a positive secret key rate. Even without any experimental noise, no secret key could be extracted at 53 km with a parameter estimation on blocks of size 1×10^6 .

Long-distance secret key generation results are shown in Fig. 2. Secret keys were produced by the experimental system at 25 km, 53 km and 80.5 km of standard optical fibre. The key rate was computed over 24 h for all distances. A sifting procedure revealed 50% of the raw key for parameter estimation, while 50% of the optical pulses were also discarded for shot noise estimation. The fraction of light pulses effectively used for generating the key was thus 25%. Error correction was performed using low density parity check codes with a graphic processing unit (GPU) decoder (see Supplementary section, 'Post-processing performance', for details). The obtained secret key rates are lower than those obtained with discrete-variable QKD¹². This is mainly a result of the lower clock rate of our experiment.

The results corresponding to the finite-size regime are of particular interest because of their relevance for practical applications. Indeed, obtaining an infinite precision in parameter estimation as required in the asymptotic case is, in practice, impossible. We can further elucidate these results by investigating the impact on the secret key rate of the uncertainty on the excess noise value. Figure 3 shows the experimental excess noise measured on blocks of size 1×10^8 over 24 h for a distance of 53 km. For each data point, a worst-case estimator of the excess noise compatible with the experimental data is also indicated. For comparison, the worst-case estimator for a block size of 1×10^6 is displayed and is clearly incompatible with the extraction of a secret key rate, thus showing that a very large block size is required to achieve long-distance QKD. These results illustrate the significance of the excess noise estimation for system performance. They also confirm the excellent stability of our system, because the excess noise maintains low values, even in this very low SNR regime required by the security proof, and with very large data blocks.

To conclude, let us discuss possible further improvements to our implementation. The current repetition rate of 1 MHz can be increased by shortening the pulse duration and the time-multiplexing period, as well as the homodyne detection data sampling period, using high-speed and high-precision data acquisition cards. Current error-correction techniques can deal with raw key rates of up to 10 Mbits s^{-1} , and better rates are possible using multiple devices. However, both the sifting procedure and the multidimensional reconciliation scheme require a large amount of classical data to be transmitted between Alice and Bob, so increasing the optical rate too much would lead to network link saturation. Finally, the ultimate secure distance that can be reached by our system is determined by the excess noise present in the setup. In this respect, recent protocols using 'noiseless amplification'²⁷ or its 'virtual' implementation^{28,29} might be promising.

Methods

Experimental details. Our experiment is a one-way implementation, where Alice sends Bob coherent light pulses with a 100 ns duration and 1 MHz repetition rate generated by a 1,550 nm pulsed telecom laser diode. These pulses are split into a weak signal and a strong LO with an unbalanced coupler. The implemented protocol uses Gaussian modulation of coherent states²⁰, whereby the signal is randomly modulated in both quadratures using amplitude and phase modulators. The signal pulses are then attenuated by a variable attenuator such that the signal power falls within a range allowing control of the variance of the Gaussian distribution exiting Alice's device using a photodiode and an appropriate feedback algorithm. A second variable attenuator lowers the signal level to a few shot noise units.

The signal and LO are then transmitted through the optical fibre without overlap using time and polarization multiplexing. Delay lines of 200 ns, composed of a 20 m single-mode fibre followed by a Faraday mirror, are used for the time multiplexing. Polarization multiplexing is achieved using polarization beamsplitters. After demultiplexing, the signal and LO interfere on a shot-noise-limited balanced pulsed homodyne detector. The electric signal coming from the detector is proportional to the signal quadrature X_ϕ , where ϕ is the relative phase between the signal and the LO, which can be controlled using the phase modulator on Bob's LO path according to the Gaussian protocol²⁰.

Feedback controls are implemented to allow for stable operation of the system over a large number of pulses ($\geq 1 \times 10^8$). Polarization drifts occurring in the quantum channel are corrected using a dynamic polarization controller. The beamsplitter placed at the entrance of Bob's apparatus aims to generate, from the LO pulse, a clock signal that is independent of polarization state. The homodyne detection statistics and an appropriate algorithm then allow us to maintain an optimal polarization state at the output of the channel. The photodiode on Alice's signal path is used for amplitude modulator feedback to correct alterations of the required voltage settings induced by temperature variations. On Bob's side, the homodyne detection output is sensitive to phase and can be used to control Alice's and Bob's phase modulators.

Security conditions. Collective attacks against the implemented protocol have been shown to be asymptotically optimal, thanks to an infinite-dimensional version of de Finetti's theorem²³. Furthermore, security proofs combining arbitrary attacks and finite-size effects are presently under active study^{24,30}.

The implemented protocol requires an exact Gaussian modulation, which is impossible to achieve. In practice, this is approximated by a truncated discretized modulation with parameters compatible with a security proof against collective attacks. This is performed with almost optimal randomness consumption using source coding techniques.

The efficiency and variance of the electronic noise of the homodyne detection are assumed to be calibrated in a secure laboratory. This corresponds to the standard realistic assumption for CVQKD implementations, according to which Eve cannot entangle herself with the losses or electronic noise of the homodyne detection. Under this assumption, we evaluate confidence intervals for these values and we compute the eavesdropper's corresponding information taking calibrated value uncertainties into account¹⁹. Note that in the so-called 'paranoid' or 'uncalibrated-device' scenario¹, where Eve can exploit the homodyne detection parameters, no secret key would be obtained beyond 35 km.

Multidimensional reconciliation. The error-correction step is divided into two parts. First, Bob divides his data into vectors $\mathbf{y}$ of size 8 and for each, draws a binary vector $\mathbf{u}$ of the same size at random; $\mathbf{u}$ is the reference for the key after error correction. Bob then computes $\mathbf{r} = \mathbf{y} \cdot \mathbf{u}$ (where the vectors are interpreted as octonions, see ref. 25 for details) and sends it to Alice who obtains $\mathbf{v} = \mathbf{x}^{-1} \cdot \mathbf{r} = \mathbf{x}^{-1} \cdot \mathbf{y} \cdot \mathbf{u}$, which is a noisy version of Bob's binary modulated vector $\mathbf{u}$, with a noise close to a Gaussian noise. Interestingly, it can be shown that the classical data $\mathbf{r}$, available to Eve, does not leak any information about the binary vector $\mathbf{u}$ (ref. 25). The second step of the error-correction protocol consists in forming vectors of size

2^{20} on Alice's and Bob's sides (corresponding to 2^{17} pairs of such vectors $\mathbf{u}$, $\mathbf{v}$) and to use multi-edge low-density parity check (LDPC) codes to correct all the errors²⁶. The amount of data revealed during this step is subtracted from the secret information previously computed. We use GPU decoding to obtain a decoding speed compatible with real-time data processing.

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Author contributions

P.J., S.K.-J. and E.D. conceived the project and developed the experimental setup. P.J. and S.K.-J. collected and analysed the data. A.L. performed the security analysis. P.G. contributed to the initial conception of the project and provided scientific expertise. All authors contributed to writing the manuscript.

Additional information

Supplementary information is available in the online version of the paper. Reprints and permissions information is available online at www.nature.com/reprints. Correspondence and requests for materials should be addressed to P.J.

Competing financial interests

The authors declare no competing financial interests.